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AGE-DEPENDENT CHANGES IN KNEE CARTILAGE T_1 , T_2 , AND $T_{1\rho}$ SIMULTANEOUSLY MEASURED USING MRI FINGERPRINTING

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Abstract

Background: Magnetic resonance fingerprinting (MRF) techniques have been recently described for simultaneous multi-parameter cartilage mapping of the knee although investigation of their ability to detect early cartilage degeneration remains limited.

Purpose: To investigate age-dependent changes in knee cartilage T_1 , T_2 , and $T_{1\rho}$ relaxation times measured using a three-dimensional (3D) MRF sequence in healthy volunteers.

Study Type: Prospective.

Subjects: The study group consisted of 24 healthy asymptomatic human volunteers (15 males with mean age 34.9 ± 14.4 yrs and 9 females with mean age 44.5 ± 13.1 yrs)

Field Strength/Sequence: A 3.0T gradient-echo-based 3D-MRF sequence was used to simultaneously create proton density-weighted images and T_1 , T_2 , and $T_{1\rho}$ maps of knee cartilage.

Assessment: Mean global cartilage and regional cartilage (lateral femur, lateral tibia, medial femur, medial tibia, and patella) T_1 , T_2 , and $T_{1\rho}$ relaxation times of the knee were measured.

Statistical Tests: Kruskal-Wallis tests were used to compare cartilage T_1 , T_2 , and $T_{1\rho}$ relaxation times between different age groups, while Spearman correlation coefficients were used to determine the association between age and cartilage T_1 , T_2 , and $T_{1\rho}$ relaxation times. $P < 0.05$ was considered statistically significant.

Results: Higher age groups showed higher global and regional cartilage T_1 , T_2 , and $T_{1\rho}$. There was a significant difference between age groups in global cartilage T_2 and $T_{1\rho}$, but no significant difference ($p=0.13$) in global cartilage T_1 . Significant differences were also present between age groups in cartilage T_2 and $T_{1\rho}$ for medial femur cartilage and medial tibia cartilage. There were

significant moderate correlations between age and T_2 and $T_{1\rho}$ for global cartilage ($R^2=0.63-0.64$), medial femur cartilage ($R^2=0.50-0.56$), and medial tibia cartilage ($R^2=0.54-0.66$).

Conclusion: Cartilage T_2 and $T_{1\rho}$ relaxation times simultaneously measured using a 3D-MRF sequence in healthy volunteers showed age-dependent changes in knee cartilage, primarily within the medial compartment.

Keywords

Cartilage; MRI fingerprinting; knee; T_1 relaxation time; T_2 relaxation time; $T_{1\rho}$ relaxation time

INTRODUCTION

Quantitative MRI parameters have been used to evaluate articular cartilage including spin-lattice relaxation time (T_1), spin-spin relaxation time (T_2), and spin-lattice relaxation time in the rotating frame ($T_{1\rho}$) (1). Among these, T_2 and $T_{1\rho}$ relaxation times are the most widely used with the largest body of literature (1). Two recent meta-analyses have documented strong evidence that knee cartilage T_2 and $T_{1\rho}$ relaxation time measurements can reliably distinguish between subjects with and without knee osteoarthritis (2,3). Cartilage T_2 and $T_{1\rho}$ mapping of the knee is commonly performed together in OA research studies for comprehensive evaluation of cartilage composition and structure (4). $T_{1\rho}$ mapping can provide information regarding the proteoglycan content of cartilage (5), while T_2 mapping can be used to assess water content and organization of the collagen fiber network (6,7). However, the sequential acquisition of multiple quantitative cartilage imaging sequences results in long MRI examination times.

Efforts to reduce MRI examination times in OA research studies have primarily focused on accelerating quantitative MRI sequences (8–11). Multiple techniques have been used to accelerate cartilage T_2 and $T_{1\rho}$ mapping of the knee including parallel imaging (8,9) and its combination with compressed sensing (10,11). Parallel imaging is inherently limited by noise amplification at high acceleration rates (8,9). Compressed sensing can reduce scan time without an increase in image noise. However, reconstruction errors leading to image blurring and reduced detail can occur when acceleration rates are beyond the image sparsity level (10,11). Deep learning-based reconstruction methods have also been described for accelerating cartilage T_2 and $T_{1\rho}$ mapping of the knee (12,13). While preliminary results are encouraging, clinical validation of deep learning-based T_2 and $T_{1\rho}$ parameter estimations of knee cartilage has yet to be performed (12,13).

Another approach to reduce MRI examination times in OA research studies is to simultaneously measure multiple relaxation time parameters of knee cartilage through a technique known as magnetic resonance fingerprinting (MRF) (14). In MRF, imaging parameters are varied continuously throughout the acquisition, which results in a unique evolution that reflects the relaxation characteristics (i.e. T_1 , T_2 , and $T_{1\rho}$) of each voxel. A pattern recognition algorithm is then used to match the fingerprints to a predefined dictionary of predicted signal evolutions created using different combinations of T_1 , T_2 , and $T_{1\rho}$ relaxation times and the imaging parameters. The combinations of T_1 , T_2 , and $T_{1\rho}$

relaxation times with the best dictionary match of the signal evolution are assigned as the measured tissue relaxation characteristics for the voxel (14).

MRF techniques have been shown to provide robust simultaneous multi-parameter relaxation time mapping for a variety of clinical applications including brain (15), abdominal (16), prostate (17), and cardiac (18) imaging. A three-dimensional (3D) MRF sequence has recently been developed for simultaneous volumetric cartilage T_1 , T_2 , and $T_{1\rho}$ mapping of the human knee at 3.0T, which can provide complete anatomic coverage with $0.7\text{mm} \times 0.7\text{mm} \times 3.0\text{mm}$ voxel size in a 9 minute scan time (19). The sequence has been shown to provide high scan-scan repeatability and good agreement of regional knee cartilage T_1 , T_2 , and $T_{1\rho}$ relaxation time measurements of the knee in human subjects compared to conventional T_1 , T_2 , and $T_{1\rho}$ mapping methods (19). However, investigation of the ability T_1 , T_2 , and $T_{1\rho}$ relaxation time measurements simultaneously obtained using a MRF technique to detect early cartilage degeneration remains limited.

This study aimed to investigate the ability of cartilage T_1 , T_2 , and $T_{1\rho}$ relaxation time measurements simultaneously acquired using the 3D-MRF sequence to detect age-related differences in cartilage composition and structure within the knee of healthy volunteers.

METHODS

Study Group

The study was performed in compliance with Health Insurance Portability and Accountability Act (HIPAA) regulations and with approval from our Institutional Review Board. The study group consisted of 24 healthy asymptomatic volunteers who had no current or prior history of joint pain, trauma, surgery, osteoarthritis, or inflammatory, or crystalline induced arthritis. All subjects signed written informed consent.

MRF Sequence Design

The 3D-MRF sequence consisted of three components. Two fast imaging with steady-state precession (FISP) segments with variable flip angles after a 10ms adiabatic inversion pulse were used to encode T_1 and T_2 relaxation times. Two fast low-angle shot (FLASH) segments similar to the FISP segments were used to encode T_1 relaxation time and B_{1+} . Four FLASH segments with 125 excitations and variable flip angles were used to encode $T_{1\rho}$ relaxation time with each segment preceded by a $T_{1\rho}$ preparation module with different turbo spin-lock (TSL) durations (TSL = 0.5, 4.2, 9.8, and 21.8 ms) (19). The optimized TSL durations were selected based upon Cramer-Rao lower bounds (20). A balanced $T_{1\rho}$ preparation module was used to compensate for the effects of B_0 and B_1 inhomogeneity. A 3D stack of stars with golden angle increments between consecutive spokes was used for readout with Cartesian sampling used in the slice direction (19).

The 3D-MRF sequence was performed in the sagittal plane on the arbitrarily chosen right knee of all subjects on the same 3.0T scanner (MAGNETOM Prisma, Siemens Healthcare, Erlangen, Germany) using a 15-channel transmit-receive coil (Quality Electrodynamics, Mayville, OH). The subjects rested for an hour before the MRI examination. Foam padding was used to secure the knee within the coil. The 3D MRF sequence was acquired with

the following imaging parameters: repetition time (TR)=7.5ms; echo time (TE)=15ms; inversion time (TI)=15ms; TSL=0.5, 3.8, 9.8, 21.8ms; flip angle (FA)=0–60°; field of view=140mm; matrix size=192×192; slice thickness=3mm; voxel size=0.7×0.7×3.0mm; receiver bandwidth=490Hz/px; number of slices=18; and total scan time=9:00min.

MRF Dictionary Construct and Simultaneous Multi-Parameter Map Reconstruction

All algorithms were implemented in MATLAB (R2020b, The MathWorks Inc., Natick, MA). The 3D image data was separated into 18 slices using fast Fourier transform and then reconstructed using non-uniform fast Fourier transform. A dictionary of simulated MRI fingerprints based on extended phase graphs (EPG) was created for T_1 , T_2 , $T_{1\rho}$, and B_{1+} values ranging with 5% increments from 50–3000ms, 2–200ms, 2–200ms, and 30–60°, respectively. The $T_{1\rho}$ preparation module was simulated as a mono-exponential decay of magnetization. The measured fingerprints and dictionary were compressed with singular value decomposition (SVD) and matched using a maximum correlation approach to simultaneously generate proton density (PD)-weighted images and T_1 , T_2 , $T_{1\rho}$, and B_{1+} maps of knee cartilage (19).

Cartilage Quantitative MRI Analysis

Cartilage quantitative MRI analysis was performed by a research assistant (AS) with 6 years of cartilage segmentation experience under the supervision of an MRI scientist (RR) with 25 years cartilage research experience using software developed in MATLAB (R2020b, The MathWorks Inc., Natick, MA). Regions of interests were drawn around the cartilage of the lateral femur, lateral tibia, medial femur, medial tibia, and patella on each sagittal PD-weighted image of the knee of each subject to create a 3D cartilage contour of each articular surface. The cartilage contours were superimposed over the cartilage T_1 , T_2 , and $T_{1\rho}$ maps to measure the mean cartilage T_1 , T_2 , and $T_{1\rho}$ relaxation times on each articular surface. Global cartilage T_1 , T_2 , and $T_{1\rho}$ relaxation times for all articular surfaces of the knee were also calculated.

Statistical Analysis

Statistical analysis was performed using the R 3.3.1 program environment (R Foundation for Statistical Computing, Vienna, Austria) with statistical significance defined as a p-value less than 0.05. Subjects were divided into four groups based upon age range (20–30ys, 30–40ys, 40–50ys, and >50ys). Kruskal-Wallis tests were used to investigate the primary hypothesis and determine differences in global cartilage T_1 , T_2 , and $T_{1\rho}$ relaxation times of the knee between age groups. For parameters that were significantly different between age groups for global cartilage, Kruskal-Wallis tests were used to determine differences in regional cartilage relaxation times between age groups, while Mann-Whitney tests were also used to compare global and regional cartilage relaxation times between younger subjects below 40 years and older subjects over 40 years of age. Spearman correlation coefficients were used to determine the association between age and global cartilage T_1 , T_2 , and $T_{1\rho}$ relaxation times with correlations between age and regional cartilage relaxation times assessed for parameters that were significantly correlated for global cartilage. The strength of correlation coefficients were assessed according to previously described criteria (21). Mann-Whitney tests were used to determine sex-related differences in global cartilage T_1 , T_2 , and $T_{1\rho}$

relaxation times of the knee with differences in regional cartilage relaxation times assessed for parameters that were significantly correlated for global cartilage.

RESULTS

The study group consisted of 24 healthy asymptomatic volunteers, which included 15 males between 24yrs and 75yrs with a mean age of 34.9yrs and standard deviation of 14.4yrs and 9 females between 21yrs and 66yrs with mean age of 44.5yrs and standard deviation of 13.1yrs. There were 5 subjects in the 20–30ys age group, 6 subjects in the 30–40yrs age group, 7 subjects in the 40–50yrs age group, and 6 subjects in the >50yrs age group.

Figure 1 shows the box plots for global and regional cartilage T_1 , T_2 , and $T_{1\rho}$ relaxation times of the knee for subjects in the different age groups. There was a significant difference between age groups in global cartilage T_2 relaxation time and global cartilage $T_{1\rho}$ relaxation time, but no significant difference ($p=0.13$) between age groups in global cartilage T_1 relaxation time. There was a significant difference between age groups in cartilage T_2 relaxation time for medial femur cartilage and medial tibia cartilage and cartilage $T_{1\rho}$ relaxation times for medial femur cartilage and medial tibia cartilage. There was a significant difference between younger and older subjects in cartilage T_2 and $T_{1\rho}$ relaxation time for global cartilage, medial femur cartilage, and medial tibia cartilage. There was no significant differences ($p=0.06$ – 0.86) between age groups or between younger and older subjects and cartilage T_2 and $T_{1\rho}$ relaxation time on the remaining articular surfaces of the knee. Figure 2 shows an example of T_1 , T_2 , and $T_{1\rho}$ maps of knee cartilage of subjects in the different age groups.

Table 1 shows the correlation coefficients describing the relationship between age and global and regional cartilage T_1 , T_2 , and $T_{1\rho}$ relaxation time measurements of the knee for all subjects. There was a significant moderate positive correlation between age and global cartilage T_2 relaxation time ($R^2=0.63$) and global cartilage $T_{1\rho}$ relaxation time ($R^2=0.64$), but no significant correlation ($p=0.13$) between age and global cartilage T_1 relaxation time ($R^2=0.33$). For cartilage T_2 relaxation time, there was a significant moderate positive correlation with age for medial femur cartilage ($R^2=0.50$) and medial tibia cartilage ($R^2=0.54$). For cartilage $T_{1\rho}$ relaxation time, there was a significant low positive correlation with age for lateral femur cartilage ($R^2=0.46$) and a significant positive moderate correlation with age for medial femur cartilage ($R^2=0.56$) and medial tibia cartilage ($R^2=0.66$).

Figure 3 shows the box plots for global and regional cartilage T_1 , T_2 , and $T_{1\rho}$ relaxation time measurements of the knee for male and female subjects. There was no significant sex-related differences ($p=0.18$ – 0.90) between male and female subjects in global cartilage T_1 relaxation times, global cartilage T_2 relaxation time, or global cartilage $T_{1\rho}$ relaxation times. Figure 4 shows an example of T_1 , T_2 , and $T_{1\rho}$ maps of knee cartilage for male and female subjects.

DISCUSSION

This study documented age-dependent differences in cartilage T_2 and $T_{1\rho}$ relaxation times of the knee, which may be due to changes in cartilage composition and structure during

the aging process. Increased cartilage $T_{1\rho}$ relaxation times with aging likely reflects changes in proteoglycan content and structure as $T_{1\rho}$ is primarily influenced by proton exchange between water molecules and glycosaminoglycan side chains of proteoglycan macromolecules (22). Normal aging of cartilage is characterized by decreased proteoglycan content (23) with smaller and less uniform proteoglycan aggregates (24), which increases proton exchange between water molecules and glycosaminoglycan side chains and hence $T_{1\rho}$ relaxation times. Increased cartilage T_2 relaxation times with aging is likely due to changes in proteoglycan macromolecular content and structure (23,24) along with increased collagen cross-linking and denaturation (25,26), which increases the mobility of bulk water in cartilage and hence T_2 relaxation times.

This study confirmed the findings of previous studies that have documented age-related differences in cartilage $T_{1\rho}$ relaxation times of the knee. Li et al (27) investigating 17 healthy volunteers and Stahl et al (28) investigating 20 healthy volunteers and 17 subjects with knee osteoarthritis found significant moderate positive correlations ($R^2=0.46-0.58$) between age and global cartilage $T_{1\rho}$ relaxation time of the knee. Goto et al. (29) also found significant low to moderate positive correlations ($R^2=0.32-0.72$) between age and global and regional cartilage $T_{1\rho}$ relaxation times of the knee in 32 healthy female volunteers. The strongest correlations between age and cartilage $T_{1\rho}$ relaxation times were present on the medial femur cartilage and medial tibia cartilage.

Previous studies on age-related changes in cartilage T_2 relaxation times of the knee are more conflicting. A study by Mosher et al. (30) investigating 30 healthy women showed a significant correlation between age and global cartilage T_2 relaxation times for subjects older but not younger than 45 years of age. Joseph et al. (31) found a significant but weak association between age and regional cartilage T_2 relaxation times for medial femur cartilage and patellar cartilage in 481 subjects in the Osteoarthritis Initiative (OAI) reference cohort between 45 and 65 years of age. However, Li et al. (27) and Stahl et al (28) et al found no significant correlations between age and global cartilage T_2 relaxation times of the knee. Wirth et al (32) also found no significant association between age and regional cartilage T_2 relaxation times of the knee in 88 subjects in the OAI reference cohort. Differences in the findings between studies may be due to differences in subject sex and age, MRI techniques used to measure cartilage T_2 and $T_{1\rho}$ relaxation times, and methods used for cartilage quantitative MRI analysis.

This study confirmed the findings of previous studies that have failed to document sex-related differences in cartilage $T_{1\rho}$ relaxation times of the knee. Mosher et al. (33) found no significant difference between 7 males and 10 females in patellar cartilage T_2 relaxation times. Wirth et al (32) also showed no significant association between sex and regional cartilage T_2 relaxation times of the knee. There is solid evidence in the literature that females have a higher prevalence of osteoarthritis with more clinical pain and joint inflammation than males (34). However, our study and others suggest that the greater osteoarthritis burden on females is not due to baseline sex-related differences in cartilage composition and structure.

This study documented age-related differences in cartilage T_1 , T_2 , and $T_{1\rho}$ relaxation times of the knee simultaneously measured using a 3D-MRF sequence with $0.7 \times 0.7 \times 3.0$ mm voxel size and 9 minute scan time. Li et al. (35) has also described a 3D magnetization-prepared angle-modulated partitioned k-space spoiled gradient-echo snapshots (MAPSS) sequence with $0.5 \times 1.1 \times 4.0$ mm voxel size and 9:30 minute scan time for simultaneous acquisition of knee cartilage T_2 and $T_{1\rho}$ relaxation time maps through use of interchangeable T_2 and $T_{1\rho}$ preparation pulses (35). The 3D-MRF technique has the additional advantage of measuring T_1 relaxation time, which could be used to assess muscle fat content (36). Muscle fatty atrophy has been shown to be associated with knee osteoarthritis (37) and significantly correlated with knee muscle strength (38), which is an important predictor of knee osteoarthritis progression (39). Simultaneous acquisition of T_1 , T_2 , and $T_{1\rho}$ relaxation times could also be potentially used for MRI synthesis to create morphologic knee images with different tissue contrasts.

Limitations

This study included a relatively small number of subjects, and it is possible that some subjects had asymptomatic cartilage degeneration of the knee that could have influenced cartilage T_1 , T_2 , and $T_{1\rho}$ relaxation times. Furthermore, the study did not compare the ability of the 3D-MRF sequence with other cartilage T_1 , T_2 , and $T_{1\rho}$ mapping techniques for detecting age-dependent changes in cartilage composition and structure. In addition, MRF techniques may be influenced by magnetization transfer effects, which may lead to inaccurate estimations of T_1 and T_2 relaxation times (40). Finally, the study could not investigate the mechanisms responsible for changes in cartilage T_1 , T_2 , and $T_{1\rho}$ relaxation times with age due to lack of histologic correlation.

Conclusions

This study demonstrated age-related differences in cartilage T_1 , T_2 and $T_{1\rho}$ relaxation times of the knee simultaneously measured using a 3D-MRF sequence. The study is an important first step toward clinical validation of the 3D-MRF sequence as differences in cartilage composition and structure between healthy knees of different ages are more subtle and difficult to detect with quantitative MRI than differences between knees with and without osteoarthritis. However, additional future clinical validation of the 3D-MRF sequence is needed including investigation of its ability to distinguish between knees with different degrees of osteoarthritis and healthy knees with and without osteoarthritis risk factors.

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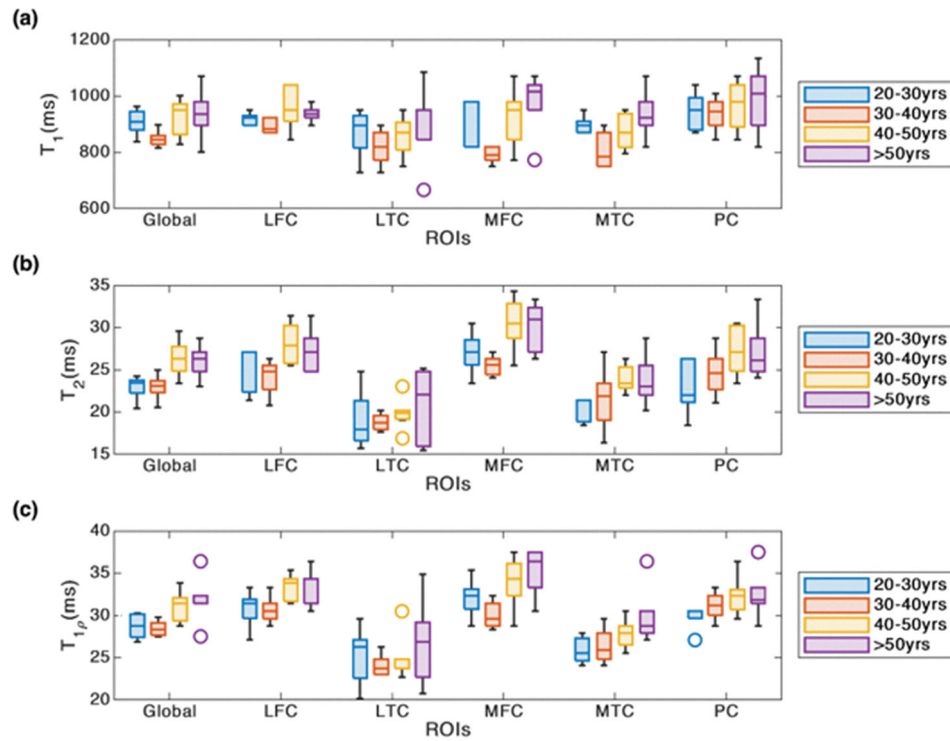


Figure 1: Box plots for global and regional cartilage T_1 , (a) T_2 (b), and $T_{1\rho}$ (c) relaxation times for subjects in the different age groups. Global=global cartilage; LFC=lateral femur cartilage; LTC=lateral tibia cartilage; MFC=medial femur cartilage; MTC=medial tibia cartilage; PC=patella cartilage.

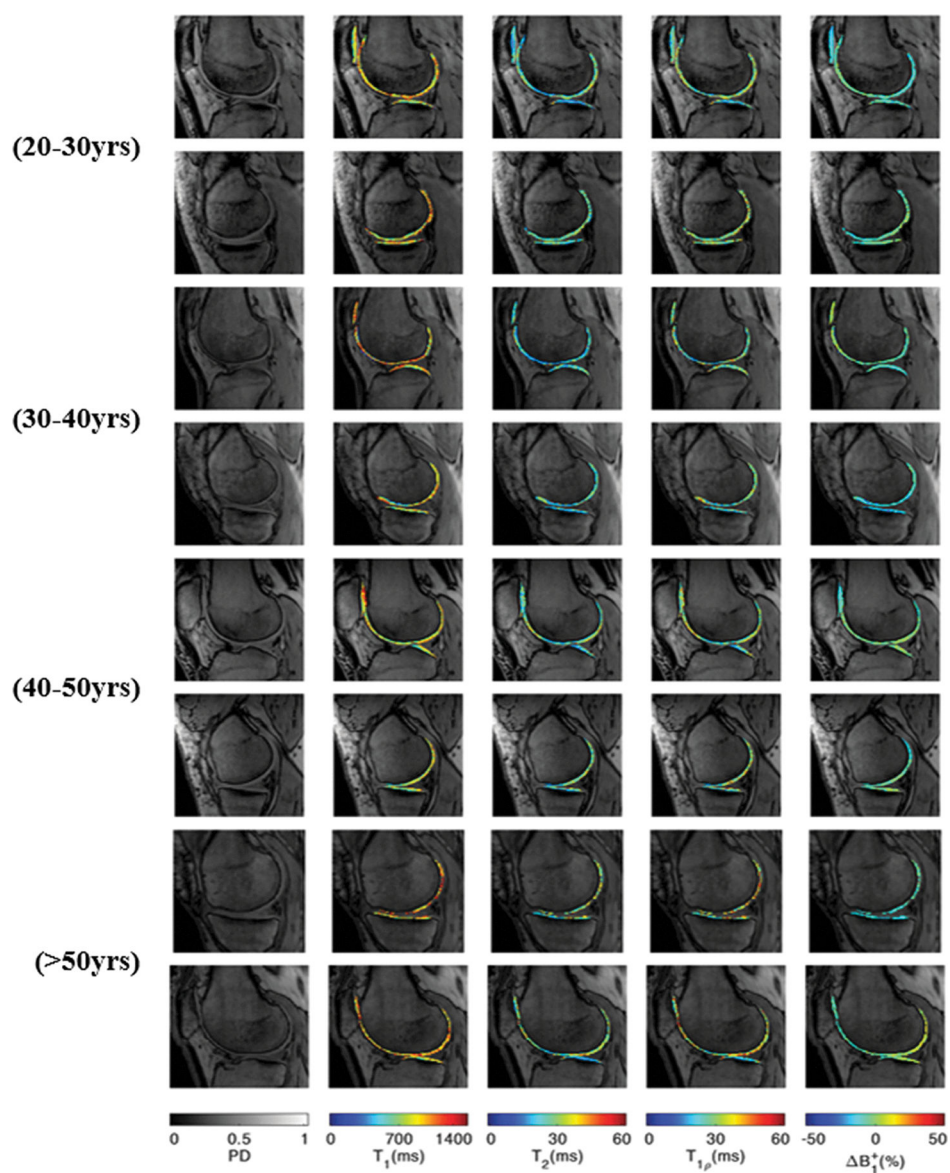


Figure 2:
Cartilage T_1 , T_2 , and $T_{1\rho}$ maps of medial femur cartilage and medial tibia cartilage of the knee for subjects in the different age groups.

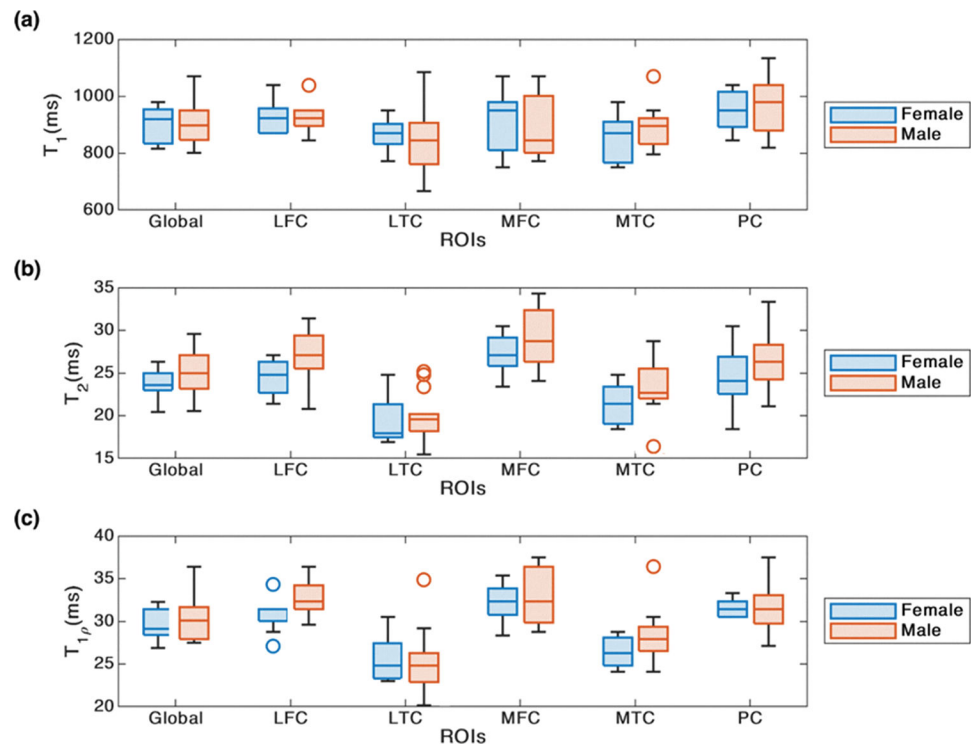


Figure 3: Box plots for global and regional cartilage T_1 , (a) T_2 (b), and $T_{1\rho}$ (c) relaxation times for male and female subjects. Global=global cartilage; LFC=lateral femur cartilage; LTC=lateral tibia cartilage; MFC=medial femur cartilage; MTC=medial tibia cartilage; PC=patella cartilage.

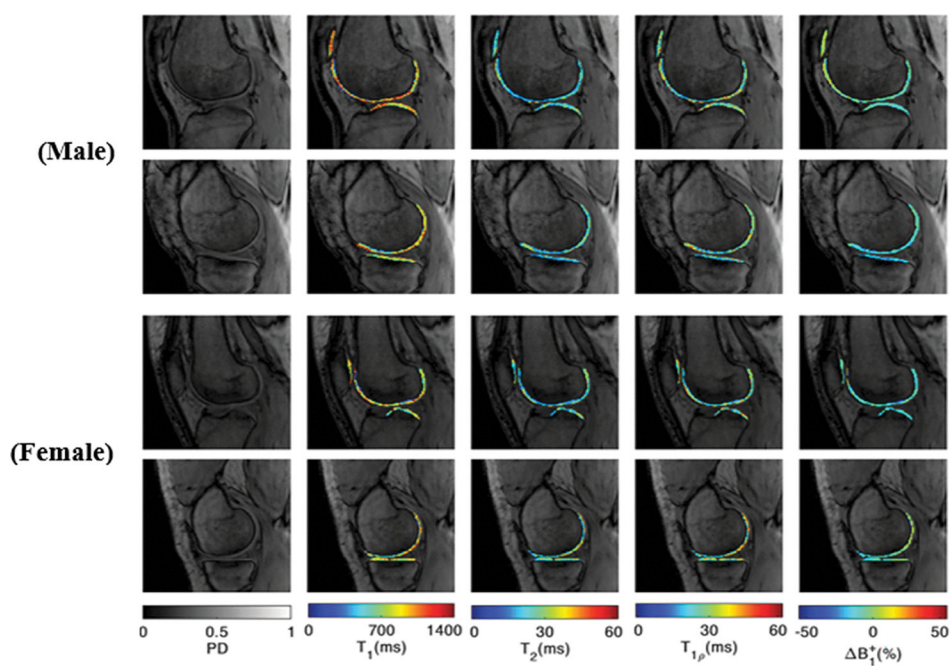


Figure 4:
Cartilage T_1 , T_2 , and $T_{1\rho}$ maps of medial femur cartilage and medial tibia cartilage of the knee for male and female subjects.

Table 1:

Spearman correlation coefficients (R^2) describing the relationship between age and global and regional cartilage T_1 , T_2 , and $T_{1\rho}$ relaxation times of the knee for all subjects. Global=global cartilage; LFC=lateral femur cartilage; LTC=lateral tibia cartilage; MFC=medial femur cartilage; MTC=medial tibia cartilage; PC=patella cartilage; N/A=non-applicable (regional cartilage correlations only performed for relaxation time parameters significantly correlated with age for global cartilage).

Surface	Cartilage Relaxation Time Parameter					
	T_1		T_2		$T_{1\rho}$	
	R^2	P-Value	R^2	P-Value	R^2	P-Value
Global	0.33	0.12	0.63	<0.01	0.64	<0.01
LFC	N/A	N/A	0.39	0.07	0.46	0.02
LTC	N/A	N/A	0.34	0.11	0.23	0.28
MFC	N/A	N/A	0.50	0.01	0.56	<0.01
MTC	N/A	N/A	0.54	0.01	0.66	<0.01
PC	N/A	N/A	0.40	0.06	0.39	0.07